INCH-POUND

MIL-DTL-28964C
04 March 1997
SUPERSEDING
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DETAIL SPECIFICATION

INSIGNIA, GRADE: ZINC

This detail specification is approved for use by the U.S. Marine Corps, Department of the Navy, and is available for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers requirements for zinc grade insignia for enlisted personnel and junior reserve officers training corps.

1.2 Classification. The insignia will be furnished in pairs in the following types and classes (see 6.2):

Type I - Enlisted Personnel
- Class 1 - Sergeant Major of the Marine Corps
- Class 2 - Sergeant Major
- Class 3 - Master Gunnery Sergeant
- Class 4 - First Sergeant
- Class 5 - Master Sergeant
- Class 6 - Gunnery Sergeant
- Class 7 - Staff Sergeant
- Class 8 - Sergeant
- Class 9 - Corporal
- Class 10 - Lance Corporal
- Class 11 - Private First Class

Type II - Junior Reserve Officers Training Corps
- Class 1 - Sergeant Major
- Class 2 - First Sergeant
- Class 3 - Gunnery Sergeant
- Class 4 - Staff Sergeant
- Class 5 - Sergeant
- Class 6 - Corporal
- Class 7 - Lance Corporal
- Class 8 - Private First Class

Beneficial comments (recommendations, additions, deletions, clarifications) and any pertinent data which may be of use in improving this document should be addressed to: Defense Personnel Support Center, Clothing and Textiles Directorate, Attn: DPSC-FNS, 2800 South 20th Street, Philadelphia, PA 19145-5099 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A **FSC 8455**

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2. APPLICABLE DOCUMENTS

2.1 Government documents

2.1.1 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

MARINE CORPS

705002A0000 - Insignia, Grade, Enlisted Personnel: Zinc.
7lC5005A0000 - Insignia, Grade, Marine Corps Junior Reserve Officers Training Corps: Zinc.

(Copies of specifications, standards, drawings and applications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicittation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) PUBLICATIONS
ASTM-B 86 - ZINC BASE ALLOY, DIE CASTING
ASTM-E 498 - METHOD OF TESTING METALS
ASTM-D 638 - METHOD OF TESTING THICKNESS

(Application for copies should be addressed to American Society for Testing and Materials, (ASTM), 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. Prior to production, the first article consisting of the number of insignias specified by the procuring activity shall be subjected to first article inspection in accordance with 4.3.

3.2 Material

3.2.1 Zinc aluminum alloy. The zinc aluminum alloy used for die casting shall conform to Alloy ASTM-B 86. The insignia shall be copper plated prior to finishing.

3.2.1.1 Thickness. The thickness for all classes of the insignia shall be 0.05 (± 0.005) inch when tested as specified in 4.4.4.

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3.2.1.2 Color and finish. The face and edges of all insignia shall be coated with a non-glossy black synthetic baking enamel finish.

3.2.2 Attaching device (prongs and clutch fasteners). Each insignia shall be provided with two (2) casted prongs having a length of .312 (± .010 inches) and two clutch fasteners conforming to drawing Nos. 705002A0000 and 71C5005A0000. The clutch fasteners shall be made from any type brass. Removal of the clutch by hand, shall not be possible without first depressing the release wings (see 4.4.1 and 4.4.7 for the required tests).

3.3 Design. The insignia shall be the Marine Corps design shown on Drawing Nos. 705002A0000 and 71C5005A0000. Each pair of type I, class 1 insignia and all classes of type II insignia shall consist of a right and left insignia. The type I, class 1 and all of type II insignia shall be produced in such a manner that when a pair of these insignia are worn, the head of the eagle or the flame end of the lamp, as applicable, shall be towards the front.

3.4 Construction. The insignia, including all center devices (i.e., Marine Corps emblem with stars, bursting bomb, single star, diamond, crossed rifles and lamp of knowledge), and zinc prongs shall be die casted and devices shall be firmly secured to the basic insignia at points of contact. All edges of insignia shall be clean, smooth and free from burrs, roughness and raggedness. The finished insignia requiring trimming shall be accurately trimmed. The prongs shall be positioned on the back of the insignia at the location shown on Drawing Nos. 705002A0000 and 7lC5005A0000. The front of the insignia shall show no signs of unevenness or distortion.

3.4.1 Direct micro-thickness. The subdued insignia is die cast from zinc aluminum specified in 3.2.1, the insignia shall be copper plated prior to finishing. When tested as specified in 4.4.3, the platen shall be a minimum of 0.0003 inch thick. When tested as specified in 4.4.4 and 4.4.5, the enamel shall not flake or chip and no color shall transfer to the cotton except for a slight trace of residue.

3.5 Finished measurement. The finished insignia shall conform to the measurements shown on Drawing Nos. 705002A0000 and 7lC5005A0000.

3.6 Marking. For Government procurements the contractor shall stamp his identification mark and the letters “GI” legibly and inconspicuously on the back or back bar of each insignia.

3.7 Workmanship. The insignia shall be accurately made and shall be clean, well finished and free from cracks, sharp or uneven edges, or any defects which might affect the appearance or serviceability. The finished insignia shall conform to the quality of product established by this document.

3.8 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

4. VERIFICATION

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2)
b. End item inspection (see 4.3)

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4.2 First article inspection. First article inspection shall be performed on one complete insignia when a first article sample is required (see 3.1). This inspection shall include the examination of 4.3 and the tests of 4.3.1. through 4.4.6.

4.3 End item inspection. The end item shall be examined for the defects listed in paragraphs 4.3.1 and 4.3.2. The sample unit shall be one pair of insignias for each type and class.

4.3.1 General defects. General defects shall be classified as follows:

Examine	Defect	Classification	
		Major	Minor
Color and finish	a. Detail altered or device does not conform to drawings	101	
	b. Surfaces not smooth	102	
	c. Discolored, stained, mottled, spotted, clearly noticeable	103	
	d. Inclusion of foreign matter or segregation's clearly noticeable	104	
	e. Abraded, scuffed, scratched or otherwise damaged	105	
	f. Paired insignia not matching in color, shade and finish (when applicable)		201
	g. Wrong color	106	
	h. Not chemically finished	107	
	i. Chemically finish is not continuous over the entire insignia	108	
Design	a. Detail altered or device does not conform to drawings	109	
	b. Any warp, twist or distortion producing irregular surface contour or outline		202
	c. Paired insignia not matching (i.e., each pair consisting of one left and one right insignia)		203
Material	Not fabricated of the specified material.	110	
Construction and workmanship	a. Nick, gouge, scratch, crack or dent		204
	b. Distorted, bent, malformed or otherwise impaired		205
	c. Surface not smooth where required, e.g., is pitted, porous or chipped	111	
	d. Edge not clean or smooth, i.e.,.fin, flash, chipped, rough, or ragged	112	
	e. Device is not centered or positioned as required	113	
	f. Device is incorrect design, missing, or not securely anchored	114	
	g. Device not die casted in one operation as required		206
	h. Insignia not accurately trimmed	115	
	i. Not pierced where required	116	
Attaching device	a. Missing	117	
	b. Defective, e.g., any part damaged or malformed, affecting use		207
	c. Not positioned as specified or not aligned perpendicular to the back	118	
	d. Prong not adequately secured to the insignia body	119	
	e. Prongs not as specified in drawing	120	
	f. Clutch will not engage prong, or is loose fitting	121	
Marking	Missing, illegible, not placed inconspicuously		208

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4.3.2 Examination of insignia for dimensions. Any dimension that is not within the tolerances specified on Drawing Nos. 705002A0000 and 7lC5005A0000 shall be classified as a defect.

4.3.3 Testing of the end item. The testing of the completely fabricated insignia shall be performed in accordance with the following for the characteristics shown therein. There shall be one determination per sample unit and results shall be reported as pass or fail. The sample unit shall be seven pair completely fabricated insignia of each type and class. The requirements are applicable to the individual unit. For the bending strength test, the sample unit shall be one item and all prongs on that item will be tested. However, one defect will be scored regardless of how many prongs fail on the item.

Characteristic	Requirement paragraph	Test method
Test for attaching device (clutches)	3.2.2	4.4.1
Test for nail strength for integral prongs.	3.2.2	4.4.7
Test for thickness	3.2.1.1	4.4.4
Test for micro-thickness	3.4.1	4.4.3
Test for bending strength and all the requirements	3.2.2	4.4.2

4.4 Tests

4.4.1 Test for clutches. The clutch shall be removed and replaced ten times from the prong. An attempt shall then be made to remove the clutch by hand without first depressing the release wings, during which time the clutch and prong shall be inspected to determine compliance with 3.2.2.

4.4.2 Test for bending strength. The prongs shall be bent five times through a 90^{o} angle, 45^{o} on each side of the prongs perpendicular position. Bending of the prongs shall be performed slowly, and each bending action shall take 10 seconds.

4.4.3 Direct micro-thickness test for copper plating. The copper plating thickness shall be determined in accordance with ASTM-E 498. The results shall be evaluated to determine compliance with 3.4.1.

4.4.4 Test for thickness. The test shall be performed in accordance with ASTM-D 638. The test shall be performed to determine compliance with 3.2.1.1.

4.4.5 Test for baked enamel. Draw a knife edge across the face side of the specimen making four parallel scores one-sixteenth inch apart. Repeat this process, drawing the knife edge this time perpendicular to the previously scored lines, making small blocks in the finish. The finish shall then be examined to determine compliance with 3.4.1.

4.4.6 Acetone test for baked enamel. The enameled surface shall be wiped five times with a piece of cotton saturated with acetone. The cotton shall be examined to determine compliance with 3.4.1.

4.4.7 Test for nail strength for integral prongs. The insignia to be tested will be placed in a jig which allows no movement of the insignia and positions prongs on a 30^{o} degree angle above the horizontal. A vertical 10 lb. load shall be applied to the top third position of each prong. No evidence of breaking or cracking shall occur in compliance with 3.2.2.

5. PACKAGING

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5.1. Packaging. For acquisition purposes, the packaging requirements shall be specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

6.1 Intended use. The insignia cover by this specification is for wear by enlisted personnel and Junior Reserve Officer's Training Corps of the Marine Corps.

6.2 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number and date of this specification.
(b) Issue of DoDISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.2.1 and 2.3)
(c) Types, classes and sizes required (see 1.2)
(d) When first article inspection is required, (see 3.1).
(e) Selection of applicable levels of preservation and packing (see 5.1).

6.3 Subject term (key word) listing.

Zinc
Insignia
R.O.T.C.

MILITARY INTERESTS:

Custodian:
Navy - MC

CIVIL AGENCY COORDINATING ACTIVITY:
GSA - FSS

PREPARING ACTIVITY:
DLA - CT

Project No. 8455-0858

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requriements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-DTL-28964	**2. DOCUMENT DATE *(YYMMDD)*** 97-03-04

3. DOCUMENT TITLE

INSIGNIA, GRADE: ZINC

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBNITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON *(If applicable)*	7. DATE SUBMITTED *(YYMMDD)*

8. PREPARING ACTIVITY

a. NAME DEFENSE PERSONNEL SUPPORT CENTER	b. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON 215-737-8105 444-8105
c. ADDRESS *(Include Zip Code)* 2800 S. 20TH STREET 12-3-C PHILADELPHIA, PA 19145	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Chruch, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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